

Identity-Based Matchmaking Encryption with Enhanced Privacy – A Generic Construction with Practical Instantiations

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Abstract. Identity-based matchmaking encryption (IB-ME), proposed by Ateniese et al. (Crypto 2019), is a type of matchmaking encryption (ME). In IB-ME, the sender can specify a target identity rcv during encryption, and the receiver can set a target identity snd during decryption. The ciphertext can be decrypted if snd matches the sender's identity σ, and rcv matches the receiver's identity ρ. The basic security notion of IB-ME is privacy, whose original definition ensures that σ, rcv, and the message remain hidden as long as $\mathsf{rcv} \neq \rho$, regardless the relation between snd and σ. Francati et al. (IndoCrypt 2021) argue that the original privacy notion is unsatisfactory as it does not match the intuitive privacy guarantee of matching encryption. They revise the original privacy notion with an *enhanced privacy* notion to characterise meaningful privacy under the condition $\mathsf{snd} \neq \sigma$ and construct an IB-ME system with the enhanced security in the plain model, albeit under a q-type pairing-based assumption. Chen et al. (AsiaCrypt 2022) leave how to construct IB-ME systems with enhanced privacy as an open problem. In this paper, we solve the problem by a generic construction of IB-ME with enhanced privacy. Instantiating our construction gives practical IB-ME systems with enhanced privacy from various standard assumptions.

Keywords: Identity-based · Matchmaking encryption · Enhanced privacy · Quantum resistance · Standard assumptions

1 Introduction

Matchmaking encryption (ME), proposed by Ateniese et al. [3], is a type of expressive encryption that allows both the sender and receiver to specify access policies so that the messages get decrypted when the policies are satisfied. Identity-based matchmaking encryption (IB-ME) is a special form of ME that supports equality policies over strings. While admitting less expressive access control policies, IB-ME, as showed by Ateniese at al. [3], is already powerful and allows interesting privacy-preserving applications, e.g., a fully anonymous bulletin board over the Tor network.

In an IB-ME system, the trusted authority issues identity-associated encryption and decryption keys. Each encryptor obtains an encryption key ek_σ on

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its identity σ. To encrypt a message m, the encryptor specifies the receiver's identity rcv and creates the ciphertext as $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{ek}_\sigma, \mathsf{rcv}, \mathsf{m})$. The decryptor whose identity is ρ obtains a decryption key dk_ρ from the authority. To decrypt ct, the decryptor sets an identity snd, and runs the decryption algorithm $\mathsf{Dec}(\mathsf{dk}_\rho, \mathsf{snd}, \mathsf{ct})$. The decryption returns the message m if the matching conditions $\rho = \mathsf{rcv}$ and $\mathsf{snd} = \sigma$ are satisfied. The security of IB-ME, addressed by a series of works [3,7,13], requires authenticity and privacy. Authenticity ensures that no one can create a valid ciphertext on an identity σ without knowing the encryption key ek_σ. Ciphertext privacy ensures that under a non-matching condition, the ciphertext ct does not reveal the underlying message *and* the identities σ and rcv.

While the authenticity definition is natural, the privacy definition is subtle. The original IB-ME privacy definition ensures that the message and the sender's identity σ remain hidden as long as the receiver's identity ρ does not match with rcv, the identity specified during encryption, regardless of the relations between snd and σ. In other words, the sender's identity σ plays no role in privacy. The original privacy definition, therefore, as argued by Francati et al. [13], does not meet the intuitive privacy guarantee of matchmaking encryption – "(the ciphertext cannot be decrypted if the) sender's attributes do not satisfy the policies held by the receiver" [3].

The generic construction of IB-ME proposed by Wang et al. [23] gives an example. In the construction, a message is encrypted using a 2-level hierarchical identity-based encryption (HIBE) with the augmented identity $\mathsf{rcv}|\sigma$. The receiver's private decryption key sk_ρ is for the higher level identity ρ. The ciphertext can be decrypted by first delegating sk_ρ to a level-2 identity key $\mathsf{sk}_{\rho|\mathsf{snd}}$ with snd chosen on the fly. Decryption works under the matching conditions, i.e., $\rho|\mathsf{snd} = \mathsf{rcv}|\sigma$. However, most HIBE systems allow using the private key of an identity to directly decrypt the ciphertext under any extended identities, e.g., sk_ρ can decrypt ciphertexts under identity $\rho|\star$, often faster than delegation. So, the receiver does not have to match the sender's identity σ for decryption. We note this is the case with lattice-based instantiation of the generic construction suggested by Wang et al. [23].

To characterise meaningful privacy for IB-ME under the mismatch condition $\mathsf{snd} \neq \sigma$ (even when $\rho = \mathsf{rcv}$) with snd chosen on the fly by any decryptor, Francati et al. [13] revise the original privacy notion with an *enhanced privacy* notion. The enhanced privacy notion ensures that the identities σ and rcv, and the underlying message from the ciphertext remain hidden, as long as one of the non-matching conditions $\sigma \neq \mathsf{snd}$ and $\rho \neq \mathsf{rcv}$ is satisfied. The model allows the adversary to get a decryption key for the identity ρ that equals rcv). Meanwhile, in IB-ME, any user, including the adversary, can generate strings snd, trying to match with the sender's identity σ. The enhanced privacy notion then requires that σ is from an (adversarially chosen) distribution with sufficient min-entropy, so a simple guessing attack does not work.

All known IB-ME systems do not meet the enhanced privacy notion except the one proposed by Francati et al. [13], which is based on Gentry's anonymous

identity-based encryption system [14] secure in the plain model under the q-ABDHE assumption, a q-type assumption from pairings. Chen et al. [7] give an IB-ME system from the standard SXDH assumption without the enhanced privacy and leave the following open problem:

> "*How can we construct IB-ME systems that satisfy the enhanced privacy under standard assumptions?*" —Chen at al. (AsiaCrypt 2022)

1.1 Our Results

We solve the problem with a generic construction of IB-ME. Our construction uses several well-studied cryptographic primitives, including an anonymous IBE system [6], an identity-based signature system [17], an average-case randomness extractor [9], and a reusable computational extractor [8].

Table 1. Comparison among IB-ME systems

IB-ME Systems	Enhanced Privacy	Assumption	ROM	Quantum Resistant
Ours + DLP'14 [11] + Cert-Falcon [12]	✔	NTRU	✔	✔
Ours + ABB'10–IBE [1] + ABB'10–IBS [1]	✔	LWE	✔	✔
Ours + BW'06 [6] + PS'06 [21]	✔	DLin+CDH	✗	✗
FGRV'21 [13]	✔	q-ABDHE	✗	✗
CLWW'22 [7]	✗	SXDH	✗	✗
AFNA'19 [3]	✗	BDH	✔	✗
WWLZ'22 [23]	✗	LWE	✔ or ✗	✔

Instantiating the primitives with existing efficient constructions from standard assumptions in the plain model leads to several concrete IB-ME systems to solve the open problem left by Chen et al. [7].

Moreover, in the random oracle model (ROM), the aforementioned primitives have practical constructions, giving practical IB-ME systems in the ROM. For example, using the practical anonymous IBE system by Ducas et al. [11] with IBS system derived from the Falcon signature [12], the certification-based approach (Scheme 1, [17]), and simple random-oracle-based (reusable) randomness extractors, we obtain the first piratical IB-ME system with enhanced privacy and quantum resistance from the NTRU assumption. Table 1 compares our instantiated IB-ME systems and the existing IB-ME systems. We discuss the instantiations of our generic IB-ME system in Sect. 5.

2 Preliminaries

Let $x \leftarrow X$ denote drawing a sample x from the distribution X. We let $x \leftarrow U(\mathcal{X})$ denote by drawing a sample x uniformly at random from a finite set $\mathcal{X}$.

Let $\mathcal{D}$ be a countable domain. The statistical distance between $X \in \mathcal{D}$ and $Y \in \mathcal{D}$ is $SD(X,Y) := \frac{1}{2}\sum_{v\in\mathcal{D}} |\Pr[X=v] - \Pr[Y=v]|$. The min-entropy of a random variable $X \in \mathcal{X}$ is $\mathrm{H}_\infty(X) := -\log(\max_x(\Pr[X=x]))$. The average min-entropy of a random variable $X \in \mathcal{X}$ given a random variable $Y \in \mathcal{Y}$ is $\tilde{\mathrm{H}}_\infty(X|Y) := -\log\left(\mathbb{E}_{y\leftarrow Y}[2^{-\mathrm{H}_\infty(X|Y=y)}]\right)$. We need the following useful lemma.

Lemma 1 ([9] Lemma 2.2). *Let X, Y and Z be random variables where Z has at most 2^λ positive-probability values. Then $\tilde{\mathrm{H}}_\infty(X|Y,Z) \geq \tilde{\mathrm{H}}_\infty(X|Y)$, and in particular, $\tilde{\mathrm{H}}_\infty(X|Z) \geq \mathrm{H}_\infty(X) - \lambda$.*

We abbreviate "probabilistic polynomial time" as "p.p.t". We denote by $\mathsf{negl}(n)$ some negligible function in n, i.e., it decreases faster than the inverse of any polynomial function. We use collision-resistant hash functions defined as follows.

Definition 1. *Let λ be the security parameter, $\mathcal{X}$, $\mathcal{Y}$ be two sets. A hash function $H : \mathcal{X} \to \mathcal{Y}$ is collision-resistant if for all p.p.t adversary $\mathcal{A}$, $\mathsf{Adv}^{\mathsf{coll}}_{H,\mathcal{A}}(\lambda) := \Pr[\mathcal{A}(\lambda, H) \to x_1, x_2 \in \mathcal{X} : x_1 \neq x_2 \wedge H(x_1) = H(x_2)]$ is negligible in λ.*

2.1 Randomness Extractors

We use information-theoretical average-case randomness extractors [9] and computational reusable randomness extractors [8].

Definition 2 (Average-Case Randomness Extractor). *A p.p.t commutable function $\mathsf{Ext} : \{0,1\}^v \times \{0,1\}^\ell \to \{0,1\}^d$ is an average-case (ℓ, k, d, ϵ)-randomness extractor if for all random variables (X,Y) with $X \in \{0,1\}^\ell$, $\tilde{\mathrm{H}}_\infty(X|Y) \geq k$,*

$$SD((K, \mathsf{Ext}(K,X), Y), (K, U(\{0,1\}^d), Y)) \leq \epsilon$$

where $K \leftarrow U(\{0,1\}^v)$.

Definition 3 (Computational Reusable Extractor). *Let λ be the security parameter. A p.p.t algorithm $\mathsf{ReExt} : \mathcal{X} \times \mathcal{K} \to \mathcal{Y}$ is a (k,m)-reusable extractor for $m = \mathsf{poly}(\lambda)$ if for all efficiently samplable random variables $\Sigma \in \mathcal{K}$ and Aux such that $\tilde{\mathrm{H}}_\infty(\Sigma|\mathsf{Aux}) \geq k$, the advantage $\mathsf{Adv}^{\mathsf{ReExt}}_{\mathcal{A}}(\lambda)$, defined as*

$$\left|\Pr[\mathcal{A}(\lambda, \mathsf{Aux}, \{(x_i, \mathsf{ReExt}(x_i,\sigma))\}_{i\in[m]}) = 1] - \Pr[\mathcal{A}(\lambda, \mathsf{Aux}, \{(x_i, y_i)\}_{i\in[m]}) = 1]\right|$$

is negligible in λ for all p.p.t adversaries $\mathcal{A}$, where $\sigma \leftarrow \Sigma$, $x_i \leftarrow U(\mathcal{X})$ and $y_i \leftarrow U(\mathcal{Y})$ are sampled independently.

2.2 Identity-Based Matchmaking Encryption

Let λ be the security parameter. An IB-ME system with identity space $\mathcal{I}$, message space $\mathcal{M}$, and ciphertext space $\mathcal{C}$ consist of five p.p.t algorithms. $\mathsf{Setup}(1^\lambda)$ returns a master public key pub and a master private key msk. We assume pub

is an implicit input to other algorithms. Given $\sigma \in \mathcal{I}$, $\mathsf{SKGen}(\mathsf{msk}, \sigma)$ returns an encryption key ek_σ. $\mathsf{RKGen}(\mathsf{msk}, \rho)$ returns a decryption key dk_ρ for $\rho \in \mathcal{I}$. Given $\mathsf{rcv} \in \mathcal{I}$ and $\mathsf{m} \in \mathcal{M}$, $\mathsf{Enc}(\mathsf{ek}_\sigma, \mathsf{rcv}, \mathsf{m})$ returns a ciphertext ct. $\mathsf{Dec}(\mathsf{dk}_\rho, \mathsf{snd}, \mathsf{ct})$ returns a message in $\mathcal{M}$ or a special symber $\perp$.

The correctness of an IB-ME system requires that if the algorithms are followed correctly, and the matching condition (i.e., $\sigma = \mathsf{snd} \wedge \rho = \mathsf{rcv}$) is satisfied, the decryption algorithm will recover the message from the ciphertext w.h.p. We follow Francati et al. [13] to define the *enhanced* privacy and authenticity.

Definition 4 (Correctness of IB-ME). *An IB-ME system* $(\mathsf{Setup}, \mathsf{SKGen}, \mathsf{RKGen}, \mathsf{Enc}, \mathsf{Dec})$ *is correct if for all* λ, $(\mathsf{pub}, \mathsf{msk}) \leftarrow \mathsf{Setup}(1^\lambda)$, σ, ρ, rcv, $\mathsf{snd} \in \mathcal{I}$, *and* $\mathsf{m} \in \mathcal{M}$ *with* $\sigma = \mathsf{snd} \wedge \rho = \mathsf{rcv}$*:*

$$\Pr\left[\mathsf{Dec}(\mathsf{dk}_\rho, \mathsf{snd}, \mathsf{ct}) = \mathsf{m}\right] \geq 1 - \mathsf{negl}(\lambda)$$

where $\mathsf{ek}_\sigma \leftarrow \mathsf{SKGen}(\mathsf{msk}, \sigma)$, $\mathsf{dk}_\rho \leftarrow \mathsf{RKGen}(\mathsf{msk}, \rho)$, *and* $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{ek}_\sigma, \mathsf{rcv}, \mathsf{m})$.

Definition 5 (Enhanced IB-ME Privacy). *Let* λ *be the security parameter. We say that an IB-ME system* Π *has privacy if for all* k*-admissible p.p.t adversary* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$*:*

$$\mathsf{Adv}^{\mathsf{priv}^+}_{\Pi,\mathcal{A}}(\lambda) := \left|\Pr\left[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1\right] - \Pr\left[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1\right]\right| \leq \mathsf{negl}(\lambda)$$

where the security experiments $\mathsf{Exp}^{\mathsf{priv}^+-b}_{\Pi,\mathcal{A}}(\lambda)$ *for* $b \in \{0,1\}$ *are defined in 1 and the probability is over the randomness of the algorithms. An adversary* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ *is* k*-admissible if* $\forall \rho \in L_2$*, it satisfies the following invariant:*

$$\begin{aligned}&(\rho \neq \mathsf{rcv}^*_0 \wedge \rho \neq \mathsf{rcv}^*_1) \vee (\mathrm{H}_\infty(\mathcal{ID}_0) \wedge \mathrm{H}_\infty(\mathcal{ID}_1) \geq k)\\ \vee\ &(\rho \neq \mathsf{rcv}^*_0 \wedge \mathrm{H}_\infty(\mathcal{ID}_1) \geq k) \vee (\mathrm{H}_\infty(\mathcal{ID}_0) \geq k \wedge \rho \neq \mathsf{rcv}^*_1)\end{aligned}$$

Experiment $\mathsf{Exp}^{\mathrm{priv}^+-b}_{\Pi,\mathcal{A}}(\lambda)$ for $b \in \{0,1\}$:

1. $L_1 \leftarrow \emptyset$, $L_2 \leftarrow \emptyset$, $L_3 \leftarrow \emptyset$
2. $(\mathsf{pub}, \mathsf{msk}) \leftarrow \mathsf{Setup}(1^\lambda)$
3. $\begin{pmatrix}\mathsf{m}^*_0, \mathsf{m}^*_1, \mathsf{rcv}^*_0, \mathsf{rcv}^*_1,\\ \mathcal{ID}_0, \mathcal{ID}_1, st\end{pmatrix} \leftarrow \mathcal{A}_1^{\mathcal{O}_1, \mathcal{O}_2}(\mathsf{pub})$
4. $\sigma^*_0 \leftarrow \mathcal{ID}_0$, $\sigma^*_1 \leftarrow \mathcal{ID}_1$
5. $\mathsf{ek}_{\sigma^*_b} \leftarrow \mathsf{SKGen}(\mathsf{msk}, \sigma^*_b)$, $\mathsf{ct}^* \leftarrow \mathsf{Enc}(\mathsf{ek}_{\sigma^*_b}, \mathsf{rcv}^*_b, \mathsf{m}^*_b)$
6. $b' \leftarrow \mathcal{A}_2^{\mathcal{O}_1, \mathcal{O}_2, \{\mathcal{O}_{3,\beta}\}_{\beta \in \{0,1\}}}(\mathsf{ct}^*, st)$

Oracle $\mathcal{O}_1(\sigma)$:

1. Return $\mathsf{ek}_\sigma \leftarrow \mathsf{SKGen}(\mathsf{msk}, \sigma)$
2. $L_1 = \{\sigma\} \cup L_1$

Oracle $\mathcal{O}_2(\rho)$:

1. Return $\mathsf{dk}_\rho \leftarrow \mathsf{RKGen}(\mathsf{msk}, \rho)$
2. $L_2 = \{\rho\} \cup L_2$

Oracle $\mathcal{O}_{3,\beta}(\mathsf{rcv}, \mathsf{m})$:

1. Return $\mathsf{Enc}(\mathsf{ek}_{\sigma^*_\beta}, \mathsf{rcv}, \mathsf{m})$
2. $L_3 = \{(\mathsf{rcv}, \mathsf{m}, \beta)\} \cup L_3$

Fig. 1. IB-ME Enhanced Privacy Experiments

Definition 6 (IB-ME Authenticity). *Let λ be the security parameter. Consider the security experiment $\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1$ defined in Fig. 2. We say that an IB-ME system Π has authenticity if for all p.p.t adversary $\mathcal{A}$:*

$$\mathsf{Adv}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) := \Pr\left[\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1\right] \leq \mathsf{negl}(\lambda)$$

where the probability is over the randomness of $\mathcal{A}$ and the IB-ME system.

Experiment $\mathsf{Exp}^{\mathrm{auth}}_{\Pi,\mathcal{A}}(\lambda)$:

1. $L_1 \leftarrow \emptyset$, $L_2 \leftarrow \emptyset$, $L_3 \leftarrow \emptyset$
2. $(\mathsf{pub}, \mathsf{msk}) \leftarrow \mathsf{Setup}(1^\lambda)$
3. $(\mathsf{ct}^*, \rho^*, \mathsf{snd}^*) \leftarrow \mathcal{A}_1^{\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3}(\mathsf{pub})$
4. $\mathsf{dk}_{\rho^*} \leftarrow \mathsf{RKGen}(\mathsf{msk}, \rho^*)$
5. $\mathsf{m}^* \leftarrow \mathsf{Dec}(\mathsf{dk}_{\rho^*}, \mathsf{snd}^*, \mathsf{ct}^*)$
6. If $\forall \sigma \in L_1 : (\sigma \neq \mathsf{snd}^*) \wedge (\mathsf{m}^* \neq \perp)$ and $(\rho^*, \mathsf{rcv}^*, \mathsf{m}^*) \notin L_3$, return 1; Otherwise, return 0

Oracle $\mathcal{O}_1(\sigma)$:

1. Return $\mathsf{ek}_\sigma \leftarrow \mathsf{SKGen}(\mathsf{msk}, \rho)$
2. $L_1 = \{\sigma\} \cup L_1$

Oracle $\mathcal{O}_2(\rho)$:

1. $\mathsf{dk}_\rho \leftarrow \mathsf{RKGen}(\mathsf{msk}, \rho)$
2. $L_2 = \{\rho\} \cup L_2$

Oracle $\mathcal{O}_3(\sigma, \mathsf{rcv}, \mathsf{m})$:

1. $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{ek}_\sigma, \mathsf{rcv}, \mathsf{m})$
2. $L_3 = \{(\sigma, \mathsf{rcv}, \mathsf{m})\} \cup L_3$

Fig. 2. IB-ME Authenticity Experiment

3 Construction

Let $\lambda \in \mathbb{N}$ be the security parameter. All parameters are functions of λ. Our construction uses the following components:

- An IBE system IBE with identity space $\mathcal{I} = \{0,1\}^\ell$, message space $\mathcal{M}_{\mathsf{ibe}}$, and ciphertext space $\mathcal{C}_{\mathsf{ibe}}$; An IBS system IBS with identity space $\mathcal{I} = \{0,1\}^\ell$ and signature space $\{0,1\}^d$. IBS is η-identity lossy.
- A collision-resistant hash function $H : \{0,1\}^* \rightarrow \{0,1\}^t$.
- A randomness extractor $\mathsf{Ext} : \{0,1\}^v \times \{0,1\}^\ell \rightarrow \{0,1\}^d$; A reusable extractor $\mathsf{ReExt} : \{0,1\}^s \times \{0,1\}^\ell \rightarrow \mathcal{C}_{\mathsf{ibe}}$.

We refer to Appendix A for the definitions of anonymous IBE and IBS, including their correctness and security.

$\mathsf{Setup}(1^\lambda)$:

1. $(\mathsf{IBE.pub}, \mathsf{IBE.msk}) \leftarrow \mathsf{IBE.Setup}(1^\lambda)$, $(\mathsf{IBS.pub}, \mathsf{IBS.msk}) \leftarrow \mathsf{IBS.Setup}(1^\lambda)$
2. $\mathsf{pub} := (\mathsf{IBE.pub}, \mathsf{IBS.pub})$, $\mathsf{msk} := (\mathsf{IBE.msk}, \mathsf{IBS.msk})$

$\mathsf{SKGen}(\mathsf{msk}, \sigma)$:

1. $\mathsf{sk}_\sigma \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \sigma)$, $\mathsf{ek}_\sigma := (\mathsf{sk}_\sigma, \sigma)$

$\mathsf{RKGen}(\mathsf{msk}, \rho)$:

1. $\mathsf{sk}_\rho \leftarrow \mathsf{IBE.Extract}(\mathsf{IBE.msk}, \rho)$, $\mathsf{dk}_\rho := (\mathsf{sk}_\rho, \rho)$

$\mathsf{Enc}(\mathsf{ek}_\sigma, \mathsf{rcv}, \sigma, \mathsf{m})$:

1. $r \leftarrow U(\{0,1\}^\ell)$, $K_1 \leftarrow U(\{0,1\}^s)$, $K_2 \leftarrow U(\{0,1\}^v)$
2. $\kappa_1 \leftarrow \mathsf{ReExt}(K_1, \sigma)$, $\kappa_2 \leftarrow \mathsf{Ext}(K_2, r)$
3. $\tilde{c} \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}, (\mathsf{m}||r))$, $c \leftarrow \tilde{c} \oplus \kappa_1$
4. $\tilde{s} \leftarrow \mathsf{IBS.Sign}\,(\mathsf{sk}_\sigma, H(c||r||K_1||K_2))$, $s \leftarrow \tilde{s} \oplus \kappa_2$
5. Return $\mathsf{ct} := (c, s, K_1, K_2)$

$\mathsf{Dec}(\mathsf{dk}_\rho, \mathsf{snd}, \mathsf{ct})$:

1. Parse ct into (c, s, K_1, K_2), return $\perp$ if ct doesn't parse
2. $\kappa_1' \leftarrow \mathsf{ReExt}(K_1, \mathsf{snd})$, $\mathsf{m}'||r' \leftarrow \mathsf{IBE.Dec}\,(\mathsf{dk}_\rho, c \oplus \kappa_1'))$
3. Return m' if $1 \leftarrow \mathsf{IBS.Ver}\,(\mathsf{snd}, H(c||r'||K_1||K_2), s \oplus \mathsf{Ext}(K_2, r'))$; Otherwise, return $\perp$

Correctness. We show that under the matching condition, i.e., $\sigma = \mathsf{snd} \wedge \rho = \mathsf{rcv}$, the original message will be recovered, i.e., $\mathsf{m}' = \mathsf{m}$. The condition $\sigma = \mathsf{snd}$ ensures $\kappa_1 = \mathsf{ReExt}(K_1, \sigma) = \mathsf{ReExt}(K_1, \mathsf{snd}) = \kappa_1'$. Under the condition $\rho = \mathsf{rcv}$ and the correctness of the IBE system IBE, with all but negligible probability, step 2 of the decryption algorithm $\mathsf{Dec}(\mathsf{dk}_\rho, \mathsf{snd}, \mathsf{ct})$ gives

$$\begin{aligned}\mathsf{m}'||r' &= \mathsf{IBE.Dec}\,(\mathsf{dk}_\rho, c \oplus \kappa_1')) = \mathsf{IBE.Dec}\,(\mathsf{dk}_\rho, c \oplus \kappa_1)) \\ &= \mathsf{IBE.Dec}\,(\mathsf{dk}_\rho, \tilde{c}) = \mathsf{m}||r\end{aligned}$$

Moreover, under the condition $\sigma = \mathsf{snd}$ and the correctness of the IBS system IBS, step 3 of the decryption algorithm returns

$$\begin{aligned}&\mathsf{IBS.Ver}\,(\mathsf{snd}, H(c||r'||K_1||K_2), s \oplus \mathsf{Ext}(K_2, r')) \\ =\ &\mathsf{IBS.Ver}\,(\sigma, H(c||r||K_1||K_2), s \oplus \mathsf{Ext}(K_2, r)) \\ =\ &\mathsf{IBS.Ver}\,(\sigma, H(c||r||K_1||K_2), \mathsf{IBS.Sign}\,(\mathsf{sk}_\sigma, H(c||r||K_1||K_2))) = 1\end{aligned}$$

with all but negligible probability. Hence, the decryption algorithm finally outputs the encrypted message m with all but negligible probability.

4 Security of the IB-ME System

The enhanced privacy (5) and authenticity (Definition 6) of the proposed generic construction can be stated by the following theorems. We refer to Appendix A for the security notions of anonymous IBE and IBS.

Theorem 1. *Let λ be the security parameter. If IBE has INDr-ID-CPA security (Definition 8), IBS is η-identity lossy (Definition 11), and ReExt is a $(k', \mathsf{poly}(\lambda))$-reusable extractor (Definition 3), and Ext is a $(\ell, k, d, \mathsf{negl}(\lambda))$-strong extractor with $k \geq \ell - t$ (Definition 2), the IB-ME system Π has the enhanced privacy (Definition 5) for all p.p.t $(k' + \eta)$-admissible adversaries.*

Theorem 2. *If IBS is EUF-ID-CMA secure (Definition 10) and hash function H is collision-resistant (Definition 1), the IB-ME system Π has authenticity.*

4.1 Proof of Enhanced Privacy

Proof (Proof of Theorem 1). In order to be valid, i.e., $k'+\eta$-admissible, according to Definition 5, the adversary $\mathcal{A}$ must satisfy one of the four mismatch conditions:

$$\bar{\mathsf{M}}_1 : \forall \rho \in L_2, \rho \neq \mathsf{rcv}_0^* \wedge \rho \neq \mathsf{rcv}_1^*$$
$$\bar{\mathsf{M}}_2 : (\mathrm{H}_\infty(\mathcal{ID}_0), \mathrm{H}_\infty(\mathcal{ID}_1)) \geq k' + \eta$$
$$\bar{\mathsf{M}}_3 : \mathrm{H}_\infty(\mathcal{ID}_0) \geq k' + \eta \wedge \forall \rho \in L_2, \rho \neq \mathsf{rcv}_1^*$$
$$\bar{\mathsf{M}}_4 : \forall \rho \in L_2, \rho \neq \mathsf{rcv}_0^* \wedge \mathrm{H}_\infty(\mathcal{ID}_1) \geq k' + \eta$$

We prove Lemmas 2, 3, 4, and 5 to prove the theorem. We use sequences of hybrid experiments. Each of the hybrid experiments outputs a binary value. We denote by $\mathsf{Hyb}_i \Rightarrow \mu$ that the i-th hybrid experiment outputs a bit value μ.

Lemma 2. *Under the condition* $\bar{\mathsf{M}}_1$*, for some negligible* $\mathsf{negl}_1(\lambda)$*, we have*

$$\left|\Pr[\mathsf{Exp}_{\Pi,\mathcal{A}}^{\mathsf{priv}^+-0}(\lambda) = 1|\bar{\mathsf{M}}_1] - \Pr[\mathsf{Exp}_{\Pi,\mathcal{A}}^{\mathsf{priv}^+-1}(\lambda) = 1|\bar{\mathsf{M}}_1]\right| \leq 4\mathsf{Adv}_{\mathsf{IBE},\mathcal{B}}^{\mathsf{indr}}(\lambda) + \mathsf{negl}_1(\lambda)$$

Proof. We define the hybrid experiments below under the condition $\bar{\mathsf{M}}_1 : \forall \rho \in L_2, \rho \neq \mathsf{rcv}_0^* \wedge \rho \neq \mathsf{rcv}_1^*$. We assume that in all hybrid experiments, for $i = 0, 1$, $\mathcal{ID}_i$ are constant distributions, so the adversary knows σ_i^*.

Hyb_0**:** Identical to $\mathsf{Exp}_{\Pi,\mathcal{A}}^{\mathsf{priv}^+-0}(\lambda)$ defined in Definition 5. On receiving $(\mathsf{m}_0^*, \mathsf{m}_1^*, \mathsf{rcv}_0^*, \mathsf{rcv}_1^*, \mathcal{ID}_0, \mathcal{ID}_1, st)$ from the adversary $\mathcal{A}_1$, the challenge ciphertext ct^* is constructed as follows:

1. For $i = 0, 1$, $\sigma_i^* \leftarrow \mathcal{ID}_i$, $\mathsf{ek}_{\sigma_i^*} \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \sigma_i^*)$
2. $r^* \leftarrow U(\{0,1\}^\ell)$, $K_1^* \leftarrow U(\{0,1\}^s)$, $K_2^* \leftarrow U(\{0,1\}^v)$
3. $\kappa_1^* \leftarrow \mathsf{ReExt}(K_1^*, \sigma_0^*))$, $\kappa_2 \leftarrow \mathsf{Ext}(K_2^*, r^*)$
4. $\tilde{c}^* \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_0^*, (\mathsf{m}_0^*||r^*))$, $c^* \leftarrow \tilde{c}^* \oplus \kappa_1^*$
5. $\tilde{s} \leftarrow \mathsf{IBS.Sign}\,(\mathsf{sk}_\sigma, H(c^*||r^*||K_1^*||K_2^*))$, $s^* \leftarrow \tilde{s}^* \oplus \kappa_2^*$
6. Return $\mathsf{ct} := (c^*, s^*, K_1^*, K_2^*)$

Hyb_1**:** This is identical to Hyb_0 except that the component $\tilde{c}^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$.

Hyb_2**:** Same as Hyb_1 except $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_0^*}, H(c^*||r_0^*||K_1^*||K_2^*)\right) \oplus w$ for a random string $w \leftarrow U(\{0,1\}^d)$.

Hyb_3**:** Same as Hyb_2 except that s^* is generated using IBS with σ_1^*, $\mathsf{ek}_{\sigma_1^*}$, and r_1^*, i.e., $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_1^*}, H(c^*||r^*||K_1^*||K_2^*)\right) \oplus w$.

Hyb_4**:** Same as Hyb_3 except $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_1^*}, H(c^*||r^*||K_1^*||K_2^*)\right) \oplus \mathsf{Ext}(K_2^*, r_1^*)$.

Hyb_5**:** Same as Hyb_4 except that κ_1^* is computed using σ_1^*, i.e., $\kappa_1^* \leftarrow \mathsf{ReExt}(K_1^*, \sigma_1^*)$.

Hyb_6**:** Same as Hyb_5 except $\tilde{c}^* \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_1^*, (\mathsf{m}_1^*||r^*))$.

The difference between Hyb_0 and Hyb_1 is how $\tilde{c}$ is computed. In Hyb_0, $\tilde{c} \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_o^*, (\mathsf{m}_0^*||r^*))$ and in Hyb_1, $\tilde{c} \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$. Under the condition $\bar{\mathsf{M}}_1$, rcv_0^* is not known to the adversary. By the INDr-ID-CPA security of IBE, a routine reduction gives an algorithm $\mathcal{B}$ (see Sect. B, Appendix) such that

$$|\Pr[\mathsf{Hyb}_0 \Rightarrow 1] - \Pr[\mathsf{Hyb}_1 \Rightarrow 1]| \leq 2\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{B}}(\lambda) \quad (1)$$

Hyb_2 differs from Hyb_1 by using a random d-bit string w to replace $\mathsf{Ext}(K_2^*, r^*)$. In Hyb_1, r^* is only used to compute s^*. Note that $H(c^*||r^*||K_1^*||k_2^*) = h$ contains at most 2^t values. Hence, as per Lemma 1, $\tilde{\mathrm{H}}_\infty(r_0^*|h) \geq \mathrm{H}_\infty(r_0^*) - t = \ell - t$. Since Ext is an $(\ell, k \geq \ell - t, d, \mathsf{negl}'_1(\lambda))$-randomness extractor, $SD((K_2^*, \mathsf{Ext}(K_2^*, r_0^*), (K_2^*, w)) \leq \mathsf{negl}'_1(\lambda)$, and thus,

$$|\Pr[\mathsf{Hyb}_1 \Rightarrow 1] - \Pr[\mathsf{Hyb}_2 \Rightarrow 1]| \leq \mathsf{negl}'_1(\lambda) \quad (2)$$

The change from Hyb_2 and Hyb_3 does not change the distribution of s^* as it is uniformly random over $\{0,1\}^d$ due to the one-time pad w. Hence,

$$\Pr[\mathsf{Hyb}_2 \Rightarrow 1] = \Pr[\mathsf{Hyb}_3 \Rightarrow 1] \quad (3)$$

Hyb_4 differs from Hyb_3 by using $\mathsf{Ext}(K_2^*, r^*)$ instead of a random string w. Using the same argument as in the proof of Inequality 2, we have

$$|\Pr[\mathsf{Hyb}_3 \Rightarrow 1] - \Pr[\mathsf{Hyb}_4 \Rightarrow 1]| \leq \mathsf{negl}''_1(\lambda) \quad (4)$$

Hyb_5 differs from Hyb_4 on computing κ_1^* using σ_1^*. This does not change the distribution of the challenge ciphertext because κ_1^* can only affect c^*, which, is random and independent of σ_0^* due to the one-time pad $\tilde{c}^*$. So, we have

$$\Pr[\mathsf{Hyb}_4 \Rightarrow 1] - \Pr[\mathsf{Hyb}_5 \Rightarrow 1]| \leq \mathsf{negl}'''_1(\lambda) \quad (5)$$

Hyb_6 sets $\tilde{c}^* := \mathsf{IBE.Enc}(\mathsf{rcv}_1^*, (\mathsf{m}_1^*||r^*))$ instead of $\tilde{c}^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$ as in Hyb_5. Using the same proof for Inequality 1 gives

$$|\Pr[\mathsf{Hyb}_5 \Rightarrow 1] - \Pr[\mathsf{Hyb}_6 \Rightarrow 1]| \leq 2\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{C}}(\lambda) \quad (6)$$

As Hyb_6 is identical to $\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda)$, under the condition $\bar{\mathsf{M}}_1$, we conclude with

$$\begin{aligned}&\left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_1] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_1]\right| \\ &\leq |\Pr[\mathsf{Hyb}_6 \Rightarrow 1] - \Pr[\mathsf{Hyb}_0 \Rightarrow 1]| \leq 4 \cdot \mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{B}}(\lambda) + \mathsf{negl}_1(\lambda)\end{aligned}$$

Lemma 3. *Under the condition* $\bar{\mathsf{M}}_2$*, for some negligible* $\mathsf{negl}(\lambda)$*, we have*

$$\left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_2] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_2]\right| \leq \mathsf{Adv}^{\mathsf{ReExt}}_{\mathcal{D}}(\lambda) + \mathsf{negl}_2(\lambda)$$

Proof. We define hybrid experiments below under the condition $\bar{\mathsf{M}}_2 : (\mathrm{H}_\infty(\mathcal{ID}_0), \mathrm{H}_\infty(\mathcal{ID}_1)) \geq k' + \eta$. The adversary does not directly know σ_0^* and σ_1^*.

Hyb_0**:** Identical to $\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda)$ (Fig. 1). On receiving $(\mathsf{m}_0^*, \mathsf{m}_1^*, \mathsf{rcv}_0^*, \mathsf{rcv}_1^*, \mathcal{ID}_0, \mathcal{ID}_1, st)$ from the adversary $\mathcal{A}_1$, the ciphertext ct^* is constructed as:

1. For $i = 0, 1$, $\sigma_i^* \leftarrow \mathcal{ID}_i$, $\mathsf{ek}_{\sigma_i^*} \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \sigma_i^*)$
2. $r^* \leftarrow U(\{0,1\}^\ell)$, $K_1^* \leftarrow U(\{0,1\}^s)$, $K_2^* \leftarrow U(\{0,1\}^v)$

3. $\kappa_1^* \leftarrow \mathsf{ReExt}(K_1^*, \sigma_0^*)$, $\kappa_2 \leftarrow \mathsf{Ext}(K_2^*, r^*)$
4. $\tilde{c}^* \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_0^*, (\mathsf{m}_0^* || r^*))$, $c^* \leftarrow \tilde{c}^* \oplus \kappa_1^*$
5. $\tilde{s} \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_\sigma, H(c^* || r^* || K_1^* || K_2^*)\right)$, $s^* \leftarrow \tilde{s}^* \oplus \kappa_2^*$
6. Return $\mathsf{ct} := (c^*, s^*, K_1^*, K_2^*)$

Hyb_1: Same as Hyb_0 except that $\kappa_1^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$ instead of $\kappa_1^* \leftarrow \mathsf{ReExt}(K_1^*, \sigma_0^*)$.

Hyb_2: Same as Hyb_1 except that $\tilde{c}^* \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_1^*, (\mathsf{m}_1^* || r^*))$.

Hyb_3: Same as Hyb_2 except that $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_0^*}, H(c^* || r^* || K_1^* || K_2^*)\right) \oplus w$ for $w \leftarrow U(\{0,1\}^d)$.

Hyb_4: Same as Hyb_3 except that s^* is generated using $\mathsf{ek}_{\sigma_1^*}$ instead of using $\mathsf{ek}_{\sigma_0^*}$, i.e., $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_1^*}, H(c^* || r^* || K_1^* || K_2^*)\right) \oplus w$ where $w \leftarrow U(\{0,1\}^d)$.

Hyb_5: Same as Hyb_4 except $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_1^*}, H(c^* || r^* || K_1^* || K_2^*)\right) \oplus \mathsf{Ext}(K_2^*, r^*)$.

Hyb_6: Same as Hyb_5 except $\kappa_1^* = \mathsf{ReExt}(K_1^*, \sigma_1^*)$.

Hyb_1 differs from Hyb_0 by setting $\kappa_1^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$. It is easy to see that oracle queries in Hyb_1 are responded in the same way as those in Hyb_0. As per Definition 5, the adversary $\mathcal{A}$ makes polynomial times, e.g., $Q_{\mathcal{O}_3}$, queries to $\mathcal{O}_{3,0}$. Assume the i-th query results in a response $\mathsf{ct}_i = (c_i, s_i, K_{1,i}, K_{2,i})$ where $c_i := \mathsf{IBE.Enc}(\mathsf{rcv}_i, (\mathsf{m}||r)) \oplus \mathsf{ReExt}(K_{1,i}, \sigma_0^*)$, $s_i := \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_0^*}, H(c_i || r_i || K_{1,i} || K_{i,2})\right) \oplus \mathsf{Ext}(K_{2,i}, r_i)$. Note that other than $\kappa_i := \mathsf{ReExt}(K_{1,i}, \sigma_0^*)$, the maximum information the queries can give about σ_0^* is $\mathsf{sk}_{\sigma_0^*}$. Since IBS is η-identity lossy,

$$\begin{aligned}\tilde{\mathrm{H}}_\infty(\sigma_0^* | \mathsf{sk}_{\sigma_0^*}, \{s_i\}_{i \in [Q_{\mathcal{O}_3}]}) &= \tilde{\mathrm{H}}_\infty(\sigma_0^* | \mathsf{sk}_{\sigma_0^*}) \geq \mathrm{H}_\infty(\sigma_0^*) - \eta \\ &= k' + \eta - \eta = k'\end{aligned}$$

Since ReExt is an $(k', \mathsf{poly}(\lambda) \geq Q_{\mathcal{O}_3} + 1)$-reusable extractor, no p.p.t adversary can tell between the distributions of κ_1^* in Hyb_0 and Hyb_1, and for some $\mathcal{D}$,

$$|\Pr[\mathsf{Hyb}_0 \Rightarrow 1] - \Pr[\mathsf{Hyb}_1 \Rightarrow 1]| \leq \mathsf{Adv}_{\mathcal{D}}^{\mathsf{ReExt}}(\lambda) \tag{7}$$

The modification from Hyb_2 does not change the adversary's view, because of the one-time pad $\kappa_1^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$, c^* is uniform over $\mathcal{C}_{\mathsf{ibe}}$, regardless of $\tilde{c}^*$. Hence

$$\Pr[\mathsf{Hyb}_1 \Rightarrow 1] = \Pr[\mathsf{Hyb}_2 \Rightarrow 1] \tag{8}$$

Hyb_3 differs from Hyb_2 by using $w \leftarrow U(\{0,1\})^d$ instead of $\mathsf{Ext}(K_2^*, r^*)$. Note that in Hyb_2, the only information about r^* for the adversary $\mathcal{A}$ is from s^* (c^* is unifromly random). Using the same argument in proving Inequality 2, we have $SD((K_2^*, \mathsf{Ext}(K_2^*, r^*), (K_2^*, w)) \leq \mathsf{negl}(\lambda)$, and thus,

$$|\Pr[\mathsf{Hyb}_2 \Rightarrow 1] - \Pr[\mathsf{Hyb}_3 \Rightarrow 1]| \leq \mathsf{negl}_2'(\lambda) \tag{9}$$

Hyb_4 differs from Hyb_3 by using $\mathsf{sk}_{\sigma_1^*}$ to compute s^*. Since s^* is uniform, so

$$\Pr[\mathsf{Hyb}_3 \Rightarrow 1] = \Pr[\mathsf{Hyb}_4 \Rightarrow 1] \tag{10}$$

Hyb_5 uses $\mathsf{Ext}(K_2^*, r^*)$ not w. Using the same argument for Inequality 9,

$$|\Pr[\mathsf{Hyb}_4 \Rightarrow 1] - \Pr[\mathsf{Hyb}_5 \Rightarrow 1]| \leq \mathsf{negl}_2''(\lambda) \tag{11}$$

The difference between Hyb_6 and Hyb_5 is similar to the difference between Hyb_1 and Hyb_0. Applying the same reasoning for Inequality 7, we have

$$|\Pr[\mathsf{Hyb}_5 \Rightarrow 1] - \Pr[\mathsf{Hyb}_6 \Rightarrow 1]| \leq \mathsf{negl}_2'''(\lambda) \tag{12}$$

Note that Hyb_6 is identical to the security experiment $\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda)$, under the condition $\bar{\mathsf{M}}_2$. Hence, $\left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_2] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_2]\right|$

$$\leq |\Pr[\mathsf{Hyb}_0 \Rightarrow 1] - \Pr[\mathsf{Hyb}_6 \Rightarrow 1]| \leq \mathsf{Adv}^{\mathsf{ReExt}}_{\mathcal{D}}(\lambda) + \mathsf{negl}_2(\lambda).$$

Lemma 4. *Under the condition* $\bar{\mathsf{M}}_3$*, for some negligible* $\mathsf{negl}(\lambda)$*, we have*

$$\left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_3] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1|\bar{\mathsf{M}}_3]\right| \leq 2\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{B}}(\lambda) + \mathsf{negl}_3(\lambda)$$

Proof. We define the hybrid experiments below under the condition $\bar{\mathsf{M}}_3 : \mathrm{H}_\infty(\mathcal{ID}_0) \geq k' + \eta \wedge \forall \rho \in L_2, \rho \neq \mathsf{rcv}_1^*$. We assume $\mathcal{ID}_1$ is a constant distribution, so the adversary knows σ_1^*. Also, the adversary can get rcv_0^* but *not* rcv_1^*.

Hyb_0 is identical to the experiment $\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda)$, as defined in Fig. 1. In particular, upon receiving $(\mathsf{m}_0^*, \mathsf{m}_1^*, \mathsf{rcv}_0^*, \mathsf{rcv}_1^*, \mathcal{ID}_0, \mathcal{ID}_1, st)$ the adversary $\mathcal{A}_1$, the challenge ciphertext ct^* is constructed as follows:

1. For $i = 0, 1$, $\sigma_i^* \leftarrow \mathcal{ID}_i$, $\mathsf{ek}_{\sigma_i^*} \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \sigma_i^*)$
2. $r^* \leftarrow U(\{0,1\}^\ell)$, $K_1^* \leftarrow U(\{0,1\}^s)$, $K_2^* \leftarrow U(\{0,1\}^v)$
3. $\kappa_1^* \leftarrow \mathsf{ReExt}(K_1^*, \sigma_0^*))$, $\kappa_2 \leftarrow \mathsf{Ext}(K_2^*, r^*)$
4. $\tilde{c}^* \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_0^*, (\mathsf{m}_0^*||r^*))$, $c^* \leftarrow \tilde{c}^* \oplus \kappa_1^*$
5. $\tilde{s} \leftarrow \mathsf{IBS.Sign}\,(\mathsf{sk}_\sigma, H(c^*||r^*||K_1^*||K_2^*))$, $s^* \leftarrow \tilde{s}^* \oplus \kappa_2^*$
6. Return $\mathsf{ct} := (c^*, s^*, K_1^*, K_2^*)$

Hyb_1: Same as Hyb_0 except that $\kappa \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$ instead $\mathsf{ReExt}(K_1^*, \sigma_0^*)$.
Hyb_2: Same as Hyb_1 except that $\tilde{c}^*$ is random from $\mathcal{C}_{\mathsf{ibe}}$, i.e., $\tilde{c}^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$.
Hyb_3: Same as Hyb_2 except that the a random d-bit string w is used for computing s^*, i.e., $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_0^*}, H(c^*||r_0^*||K_1^*||K_2^*)\right) \oplus w$.
Hyb_4: Same as Hyb_3 except that s^* is generated using $\mathsf{ek}_{\sigma_1^*}$ instead of using $\mathsf{ek}_{\sigma_0^*}$, i.e., $s^* \leftarrow \mathsf{IBS.Sign}\left(\mathsf{sk}_{\sigma_1^*}, H(c^*||r_1^*||K_1^*||K_2^*)\right) \oplus w$ where $w \leftarrow U(\{0,1\}^d)$.
Hyb_5: Same as Hyb_4 except that $\mathsf{Ext}(K_2^*, r_1^*)$ is used to compute s^*, not w.
Hyb_6: Same as Hyb_5 except that κ is computed using $\kappa \leftarrow \mathsf{ReExt}(K_1^*, \sigma_1^*)$.
Hyb_7: Same as Hyb_6 except $\tilde{c}^* \leftarrow \mathsf{IBEEnc}(\mathsf{rcv}_1^*, (\mathsf{m}_1^*||r^*))$.

The difference between Hyb_1 and Hyb_0 here (under condition $\bar{\mathsf{M}}_3$) is similar to the difference between Hyb_1 and Hyb_0 under condition $\bar{\mathsf{M}}_2$. It is easy to see that the oracle queries in Hyb_1 are responded to the same as those in Hyb_0. Applying the same proof for Inequality 7, we get, for some $\mathcal{D}$

$$|\Pr[\mathsf{Hyb}_0 \Rightarrow 1] - \Pr[\mathsf{Hyb}_1 \Rightarrow 1]| \leq \mathsf{Adv}^{\mathsf{ReExt}}_{\mathcal{D}}(\lambda) \tag{13}$$

The modification from Hyb_2 does not change the adversary's view, because of the one-time pad $\kappa_1^* \leftarrow U(\mathcal{C}_{\mathsf{ibe}})$, c^* is uniform over $\mathcal{C}_{\mathsf{ibe}}$, regardless of $\tilde{c}^*$. Hence

$$\Pr[\mathsf{Hyb}_1 \Rightarrow 1] = \Pr[\mathsf{Hyb}_2 \Rightarrow 1] \tag{14}$$

Hyb_3 uses a random d-bit string w to compute s^*, rather than $\mathsf{Ext}(K_2^*, r^*)$. Using the same proof for Inequality 9, we get

$$|\Pr[\mathsf{Hyb}_2 \Rightarrow 1] - \Pr[\mathsf{Hyb}_3 \Rightarrow 1]| \leq \mathsf{negl}_3'(\lambda) \tag{15}$$

The changes made by Hyb_4 from Hyb_3, i.e., using $\mathsf{sk}_{\sigma_1^*}$ instead of using $\mathsf{sk}_{\sigma_0^*}$ to compute s^*, does not change the distribution of s^*, because s^* is uniformly random due the one-time pad w. Therefore,

$$\Pr[\mathsf{Hyb}_3 \Rightarrow 1] = \Pr[\mathsf{Hyb}_4 \Rightarrow 1] \tag{16}$$

The difference between Hyb_5 and Hyb_4 is essentially the same as that between Hyb2 and Hyb_3. Applying the same argument in proving Inequality 15 leads to

$$|\Pr[\mathsf{Hyb}_4 \Rightarrow 1] - \Pr[\mathsf{Hyb}_5 \Rightarrow 1]| \leq \mathsf{negl}_3''(\lambda) \tag{17}$$

The changes in Hyb_6 do not modify the distribution of ct^* (in particular c^*). This is because the ciphertext component c^* is masked by $\tilde{c}^*$, which is uniformly random on $\mathcal{C}_{\mathsf{ibe}}$ in both hybrid experiments, even the adversary knows σ_1^*. So,

$$\Pr[\mathsf{Hyb}_5 \Rightarrow 1] = \Pr[\mathsf{Hyb}_6 \Rightarrow 1] \tag{18}$$

Hyb_7 changes the way of computing $\tilde{c}$ by $\tilde{c}^* \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}_1^*, (\mathsf{m}_1^* || r))$. Recall that under condition $\bar{\mathsf{M}}_3$, rcv_1^* is not known to the adversary. Under the security of IBE, a routine reduction gives

$$|\Pr[\mathsf{Hyb}_6 \Rightarrow 1] - \Pr[\mathsf{Hyb}_7 \Rightarrow 1]| \leq 2\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{B}}(\lambda) \tag{19}$$

for some INDr-ID-CPA adversary $\mathcal{B}$.

Hyb_7 is identical to the experiment $\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda)$, under the condition $\bar{\mathsf{M}}_3$. We conclude with $\left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1 | \bar{\mathsf{M}}_3] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1 | \bar{\mathsf{M}}_3]\right|$

$$\leq |\Pr[\mathsf{Hyb}_0 \Rightarrow 1] - \Pr[\mathsf{Hyb}_7 \Rightarrow 1]| \leq 2\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{B}}(\lambda) + \mathsf{Adv}^{\mathsf{ReExt}}_{\mathcal{D}}(\lambda) + \mathsf{negl}_3(\lambda)$$

Lemma 5. *Under the condition* $\bar{\mathsf{M}}_4$*, for some negligible* $\mathsf{negl}(\lambda)$*, we have*

$$\left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1 | \bar{\mathsf{M}}_3] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1 | \bar{\mathsf{M}}_3]\right| \leq 2\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{B}}(\lambda) + \mathsf{negl}_4(\lambda)$$

Proof. The proof is identical to that of Lemma 4 and thus omitted.

Combining Lemma 2, Lemma 3, Lemma 4, and Lemma 5, we have

$$\begin{aligned}
\mathsf{Adv}^{\mathsf{priv}^+}_{\Pi,\mathcal{A}}(\lambda) &= \left|\Pr\left[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1\right] - \Pr\left[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1\right]\right| \\
&\leq \sum_{j=1}^{4} \left|\Pr[\mathsf{Exp}^{\mathsf{priv}^+-0}_{\Pi,\mathcal{A}}(\lambda) = 1 | \bar{\mathsf{M}}_i] - \Pr[\mathsf{Exp}^{\mathsf{priv}^+-1}_{\Pi,\mathcal{A}}(\lambda) = 1 | \bar{\mathsf{M}}_i]\right| \\
&\leq 8 \cdot \mathsf{Adv}^{\mathsf{indr}}_{\Pi,\mathcal{B}}(\lambda) + 2 \cdot \mathsf{Adv}^{\mathsf{ReExt}}_{\mathcal{D}}(\lambda) + \mathsf{negl}(\lambda)
\end{aligned}$$

which is negligible under the hypotheses. This ends the proof of Theorem 1.

4.2 Proof of Authenticity

Proof (Proof of Theorem 2). Let $(\mathsf{ct}^*, \rho^*, \mathsf{snd}^*) \notin L_3$ where $\mathsf{ct}^* = (c^*, s^*, K_1^*, K_2^*)$ be the adversary $\mathcal{A}$'s output in $\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda)$ such that $\mathsf{m}^*||r^* \leftarrow \mathsf{IBE.Dec}(\rho^*, c^*)$. Let $(\sigma_i, \mathsf{rcv}_i, \mathsf{m}_i)$ be $\mathcal{A}$'s i-th query to the oracle $\mathcal{O}_3$, and $(c_i, s_i, K_{1,i}, K_{2,i})$ be the response where $K_{1,i}, K_{2,i}$ are random; $c_i = \mathsf{IBE.Enc}(\mathsf{rcv}_i, (m_i||r_i))$, and $s_i = \mathsf{IBS.Sign}(\mathsf{sk}_{\sigma_i}, H(c_i||r_i||K_{i,1}||K_{i,2})) \oplus \mathsf{Ext}(K_{i,2}, r_i)$. Let E be the event that there is i such that $(c^*, r^*, K_1^*, K_2^*) \neq (c_i, r_i, K_{1,i}, K_{2,i})$ but $H(c^*||r^*||K_1^*||K_2^*) = H(c_i||r_i||K_{1,i}||K_{2,i})$. It is easy to see that The probability the E happens is bounded by $\mathsf{Adv}^{\mathsf{coll}}_{H,\mathcal{B}}(\lambda)$ for some p.p.t adversary $\mathcal{B}$.

We show that if the event $\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1$ happens conditioned on $\neg E$, we can build an algorithm $\mathcal{B}'$ that breaks the EUF-ID-CMA security of IBS. The reduction between $\mathcal{B}'$ and an EUF-ID-CMA security challenger works as follows.

1. Upon receiving $\mathsf{IBS.pub}$ from the EUF-ID-CMA security challenger, $'\mathcal{B}$ generates $(\mathsf{IBE.pub}, \mathsf{IBE.msk}) \leftarrow \mathsf{IBE.Setup}(1^\lambda)$, outputs $\mathsf{pub} := (\mathsf{IBE.pub}, \mathsf{IBS.pub})$, and keeps $\mathsf{IBE.msk}$. It also initialises three lists $L_1, L_2, L_3 \leftarrow \emptyset$.
2. $\mathcal{B}'$ responds to $\mathcal{A}$'s queries as follows:
   - For an $\mathcal{O}_1$-query σ: $\mathcal{B}'$ queries the oracle $\mathsf{OExt}(\sigma)$ to get sk_σ, and forwards $\mathsf{ek}_\sigma = (\mathsf{sk}_\sigma, \sigma)$ to $\mathcal{A}$, and then sets $L_1 \leftarrow L_1 \cup \sigma$. We note that in this case, $\sigma \in L_{\mathsf{id}}$ of the IBS challenger.
   - For an $\mathcal{O}_2$-query ρ: $\mathcal{B}'$ returns $\mathsf{dk}_\rho \leftarrow \mathsf{IBE.Extract}(\mathsf{IBE.msk}, \rho)$ to $\mathcal{A}$. Then, $\mathcal{B}'$ updates the list by $L_2 \leftarrow L_2 \cup \rho$.
   - For an $\mathcal{O}_3$-query $(\sigma, \mathsf{rcv}, \mathsf{m})$: $\mathcal{B}'$ samples random r, K_1, K_2, and computes $c \leftarrow \mathsf{IBE.Enc}(\mathsf{rcv}, (\mathsf{m}||r)) \oplus \mathsf{ReExt}(K_1, \sigma)$, $\kappa_2 \leftarrow \mathsf{Ext}(K_2, r)$. Then, $\mathcal{B}'$ makes a query $(\sigma, H(c||r||K_1||K_2))$ to the oracle OSig to receive a signature $\tilde{s}$. $\mathcal{B}'$ returns the IB-ME ciphertext $(c, s \leftarrow \tilde{s} \oplus \kappa_2, K_1, K_2)$ as the response to the $\mathcal{A}$'s query and sets $L_3 \leftarrow L_3 \cup (\sigma, \mathsf{rcv}, \mathsf{m})$. We note that the tuple $(\sigma, H(c||r||K_1||K_2), \tilde{s}) \in L_s$ of the IBS challenger.
3. Finally, the adversary outputs a tuple $(\mathsf{ct}^*, \rho^*, \mathsf{snd}^*)$. $\mathcal{B}'$ proceeds as follows:
   - Parse $\mathsf{ct}^* = (c^*, s^*, K_1^*, K_2^*)$; Abort if ct^* does not parse correctly.
   - Set $\mathsf{sk}_{\rho^*} \leftarrow \mathsf{IBE.Extract}(\mathsf{IBE.msk}, \rho)$, compute $\mathsf{m}^*||r^* \leftarrow \mathsf{IBE.Dec}(\mathsf{sk}_{\rho^*}, c^* \oplus \mathsf{ReExt}(K_1^*, \mathsf{snd}^*))$.
   - Abort if $\mathsf{snd}^* \in L_1$ or $(\mathsf{snd}^*, \rho^*, \mathsf{m}^*) \in L_3$.
   - Output $(\mathsf{snd}^*, H(c^*||r^*||K_1^*||K_2^*), s^* \oplus \mathsf{Ext}(K_2^*, r^*))$ to the challenger if $\mathsf{IBS.Ver}(\mathsf{snd}^*, H(c^*||r^*||K_1^*||K_2^*), s^* \oplus \mathsf{Ext}(K_2^*, r^*)) \neq 1$.

We analyse the reduction. It is easy to see that the distribution of pub and the output distributions of the oracles $\mathcal{O}_1$, $\mathcal{O}_2$, and $\mathcal{O}_3$ simulated by $\mathcal{B}'$ are identical to those from the IB-ME authenticity security experiment. We argue that if $\mathcal{A}$'s output $(\mathsf{ct}^*, \rho^*, \mathsf{snd}^*)$ makes the experiment $\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda)$ output 1, then the output of $\mathcal{B}'$, i.e., $(\mathsf{snd}^*, H(c^*||r^*||K_1^*||K_2^*), s^* \oplus \mathsf{Ext}(K_2^*, r^*))$ gives a valid forgery, making the experiment $\mathsf{Exp}^{\mathsf{euf}}_{\mathsf{IBS},\mathcal{B}'}(\lambda)$ output 1. We prove this by contrapositive.

1. If $(\mathsf{snd}^*, H(c^*||r^*||K_1^*||K_2^*), s^* \oplus \mathsf{Ext}(K_2^*, r^*))$ cannot be verified by the IBS system IBS, then $(\mathsf{ct}^*, \rho^*, \mathsf{snd}^*)$ cannot be a valid ID-ME ciphertext.

2. If $\mathsf{snd}^* \in L_{\mathsf{id}}$, we must have $\mathsf{snd}^* \in L_1$. Therefore, $(\mathsf{ct}^*, \rho^*, \mathsf{snd}^*)$ is not a valid forgery to IB-ME authenticity.
3. Considering that the tuple $(\mathsf{snd}^*, H(c^*||r^*||K_1^*||K_2^*), s^* \oplus \mathsf{Ext}(K_2^*, r^*)) \in L_s$, i.e., the tuple was asked by $\mathcal{B}'$ to the signing oracle OSig where c^* was created by rcv^*. This must be caused by an $\mathcal{O}_3$-query $(\sigma^* = \mathsf{snd}^*, \mathsf{rcv}^* = \rho^*, \mathsf{m}^*)$, made by $\mathcal{A}$. Under the condition $\neg E$, there is no tuple $(c, r, K_1, K_2) \neq (c^*, r^*, K_1^*, K_2^*)$. By the IB-ME construction, $c^* = \mathsf{IBE.Enc}(\mathsf{rcv}^*, (\mathsf{m}^*||r^*)) \oplus \mathsf{ReExt}(K_1^*, \sigma^*)$. Since ReExt is deterministic under the inputs K_1^* and σ^*, there is no another message $\mathsf{m} \neq \mathsf{m}^*$ such that $c^* = \mathsf{IBE.Enc}(\mathsf{rcv}^*, (\mathsf{m}||r^*)) \oplus \mathsf{ReExt}(K_1^*, \sigma)$[1] and $(\mathsf{snd}^* = \sigma^*, \rho^* = \mathsf{snd}^*, \mathsf{m}) \notin L_3$. Hence, it must be the case that the invalid IBS forgery $(\mathsf{snd}^*, H(c^*||r^*||K_1^*||K_2^*), s^* \oplus \mathsf{Ext}(K_2^*, r^*))$ (as it is in L_s) links to $(\mathsf{ct}^*, \mathsf{snd}^* = \sigma^*, \rho^* = \mathsf{rcv}^*)$, an invalid IB-ME forgery due to $(\mathsf{snd}^* = \sigma^*, \rho^* = \mathsf{snd}^*, \mathsf{m}) \notin L_3$.

To sum up, conclude the proof by

$$\begin{aligned}\Pr[\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1] &= \Pr[\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1|E]\Pr[E] + \Pr[\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1|\neg E]\Pr[\neg E] \\ &\leq \Pr[E] + \Pr[\mathsf{Exp}^{\mathsf{auth}}_{\Pi,\mathcal{A}}(\lambda) = 1|\neg E] \leq \mathsf{Adv}^{\mathsf{coll}}_{H,\mathcal{B}}(\lambda) + \mathsf{Adv}^{\mathsf{euf}}_{\mathsf{IBS},\mathcal{B}'}(\lambda).\end{aligned}$$

5 Instatiations

5.1 Anonymous IBE Systems and IBS Systems

Our IB-ME construction requires the underlying anonymous IBE system to have INDr-ID-CPA security, which most anonymous IBE systems satisfy.

Table 2. Example anonymous IBE systems applicable to our IB-ME system

Schemes	Assumption	ROM	Quantum Resistant
BF'01–Basicident [5]	DBDH	✔	✗
BW'06 [6]	DLin	✗	✗
DLP'14 [11]	NTRU/RLWE	✔	✔
GPV'08 [15]	LWE	✔	✔
ABB10'10–Select-ID [1]	LWE	✗	✔

Table 2 summarises a list of anonymous IBE systems applicable to our generic construction. Some anonymous IBE systems, e.g., BW'06 system, are proven in a weaker selective identity model (INDr-sID-CPA). These systems are efficient and INDr-ID-CPA secure via complexity leveraging [4].

Identity-based signature is another well-studied primitive needed by our construction. One of the simple and effective ways to construct IBS systems is the certification-based approach (termed Cert below in Table 3). The

[1] Otherwise the ciphertext decrypts to two different messages.

certification-based approach uses a normal digital signature system. Applying the certification-based approach to practical post-quantum digital signature systems, e.g., Falcon [12] and Dilithium [10] yield efficient IBS systems comparable to the direct post-quantum IBS systems. We refer to Scheme 1, [17] for the formal descriptions of the IBS system based on the certification-based approach.

Table 3. Example IBS Systems applicable to our IB-ME system

Schemes	Assumption	ROM	Quantum Resistant
Cert+ Falcon [12]	NTRU	✔	✔
Cert+ Dilithium [10]	MLWE	✔	✔
GS'02 [16]	CDH	✔	✗
PS'06 [21]	CDH	✗	✗
ABB'10 [15]	SIS	✗	✔

Our IB-ME construction requires IBS systems to be identity lossy (Definition 11), i.e., the identity key contains less information than that of the identity if the identity is sampled uniformly at random. Without changing the IBS system construction much, identity lossiness is easy to satisfy: Assuming the identity space is large enough and a collision-resistant hash function $H : \{0,1\}^n \rightarrow \{0,1\}^m$ that compresses the inputs (note, most hash functions compress), i.e., $n > m$, we use $H(\mathsf{id})$ as the actual identity for the IBS identity key extraction and other algorithms. Since $H(\mathsf{id})$ losses information of id, the identity key $\mathsf{sk}_{\mathsf{id}}$ has at most m bits information, and the IBS system is $(n-m)$-identity lossy. Table 3 lists IBS systems applicable to our IB-ME construction.

5.2 Randomness Extractors

Our IB-ME construction uses average-case randomness extractors and reusable computational extractors. The two types of extractors have simple and practical instantiations in the ROM. Let $F : \{0,1\}^\ell \rightarrow \{0,1\}^d$ be a random oracle. We use $\mathsf{Ext} := F(\cdot)$ equivalently as the $(\ell, k, d, \mathsf{negl}(\lambda))$-randomness extractor for us whose security is based on Lemma 2, [18].

The reusable computational extractor for our IB-ME system can be constructed as $\mathsf{ReExt}(\cdot, \sigma) := G(\cdot, F'(\sigma))$ where secret seed σ is from a distribution over $\{0,1\}^\ell$ with min-entropy k, $F' : \{0,1\}^\ell \rightarrow \{0,1\}^d$ and $G : \{0,1\}^s \times \{0,1\}^d \rightarrow \mathcal{C}_{\mathsf{ibe}}$ are independent random oracles. Each randomness extraction chooses a random value $K \leftarrow U(\{0,1\}^s)$ and outputs $(K, G(K, F'(\sigma)))$. The security of the above reusable extractor follows from Lemma 2, [22].[2] Note that we use F as a randomness extractor to extract nearly uniformly random values $F'(\sigma)$ for

[2] The Lemma was originally for proofs in the Quantum Random Oracle Model. It also applies to the ROM as indicated by Saito et al.

$\mathsf{ReExt} := G(\cdot, F'(\sigma))$ so that Lemma 2, [22] applies. In practice, we can set $\mathsf{Ext} := F(\cdot)$ and $\mathsf{ReExt} := G(\cdot, F(\sigma))$ with secure hash functions, e.g., SHA-3.

In the plain model, the average-case randomness extractor can be obtained from *universal hash functions*. We refer to the work of Dodis et al. [9] for more details. If the plain model does not require quantum resistance, we can instantiate reusable computational extractors by Noar and Segev's DDH-based construction [20]. To ensure quantum resistance, we can use the reusable extractor on the Learning Parity with Noise (LPN) problem and the Learning With Errors (LWE) problem – Dodis et al. [8] design a reusable extractor from a leaky LPN assumption. Yu and Zhang [24] give direct proof that the hardness of the leaky LPN problem is polynomially equivalent to the standard LPN problem. Consequently, we can obtain a reusable extractor using Dodis et al.'s construction and base its security on the standard LPN problem. Similarly, we can use the leaky LWE problem by Alwen et al. [2], which was proved as hard as the standard LWE problem (for certain parameters), to build a reusable extractor. The extractor construction is essentially the same as Dodis et al.'s construction.

5.3 A Practical IB-ME Instantiation from NTRU

According to the discussion, we can use, for example, the Boneh-Frankin IBE system [5] and the Gentry-Silverberg IBS system [16] (obtained via the 2-level Gentry-Silverberg HIBE) to get a practical IB-ME system with enhanced privacy in the ROM, provided quantum resistance is not a concern. To get a practical quantum-resistant IB-ME system, we consider the anonymous IBE system by Ducas et al. [11] (referred to as DLP-IBE) and the IBS system from Falcon signature [12] using the certification-based approach.

For DLP-IBE, we consider the improved implementation by McCarthy et al. [19] that achieves 192-bit security (Parameter Set 3, Table 3, [19]). The implemented system has master public key size, private key size, and ciphertext size $\approx$ 2950 bytes, $\approx$ 2180 bytes, and $\approx$ 3720 bytes, respectively.

We use Falcon-1024, which achieves 256-bit security, together with the above implementation of the DLP'14 system, to provide 192-bit security (NIST security level 4). As per Table 3.3, [12], the public-key size is $\approx$ 1790 bytes, signature size is 1280 bytes. The size of the Falcon's signing key can be at most 3090 bytes as per Sect. 4.4.3, [12]. We summarise the parameters in Table 4, where $|\mathsf{pub}|$, $|\mathsf{ek}_\sigma|$, $|\mathsf{dk}_\rho|$, and $|\mathsf{ct}|$ represent the size of the master public key, the encryption key, the decryption key, the ciphertext of the IB-ME system. Computational performance-wise, DLP-IBE has an encryption time of 0.239 ms and a decryption time of 0.057 ms, and Falcon has a signing time of 0.34 ms and a verification

Table 4. Example practical IB-ME Instantiations Based on NTRU

$\lvert\mathsf{pub}\rvert$	$\lvert\mathsf{ek}_\sigma\rvert$	$\lvert\mathsf{dk}_\rho\rvert$	$\lvert\mathsf{ct}\rvert$	Assumption
$\approx$ 4740 bytes	$\approx$ 5250 bytes	$<$ 6160 bytes	$\approx$ 6530 bytes	NTRU

time of 0.07 ms (speed depends on the hardware used in [12,19]). For our IB-ME systems in the ROM, the encryption involves an IBE encryption and an IBS signing, and the decryption involves an IBE decryption and an IBS verification. The extractors' operations are hashing operations. The overall computational performance of our IB-ME system is practical.

Appendix A Identity-Based Encryption and Signature

An Identity-Based Encryption (IBE) system IBE with identity space $\mathcal{I}$ and message space $\mathcal{M}$ and ciphertext space $\mathcal{C}$ consists of four p.p.t algorithms. $\mathsf{IBE.Setup}(1^\lambda)$ returns a master public key pub and a master private key msk. We assume that pub is an implicit input to the other algorithms. $\mathsf{IBE.Extract}(\mathsf{msk}, \mathsf{id})$ generates a private key $\mathsf{sk_{id}}$. $\mathsf{IBE.Enc}\ (\mathsf{id}, \mathsf{m})$ returns a ciphertext $\mathsf{ct} \in \mathcal{C}$. $\mathsf{IBE.Dec}\ (\mathsf{sk_{id}}, \mathsf{id}, \mathsf{ct})$ returns a message $\mathsf{m} \in \mathcal{M}$ or $\perp$. We define the correctness as follows.

Definition 7. *An IBE system* ($\mathsf{IBE.Setup}$, $\mathsf{IBE.Extract}$, $\mathsf{IBE.Enc}$, $\mathsf{IBE.Dec}$) *with identity space* $\mathcal{I}$*, message space* $\mathcal{M}$*, and ciphertext space* $\mathcal{C}$ *is correct if for all* $\lambda \in \mathbb{N}$*,* $(\mathsf{pub}, \mathsf{msk}) \leftarrow \mathsf{IBE.Setup}(1^\lambda)$*,* $\mathsf{id} \in \mathcal{I}$*,* $\mathsf{m} \in \mathcal{M}$*, we have*

$$\Pr\left[\mathsf{IBE.Dec}(\mathsf{sk_{id}}, \mathsf{id}, \mathsf{Enc}(\mathsf{id}, \mathsf{m})) = \mathsf{m}\right] \geq 1 - \mathsf{negl}(\lambda)$$

where $\mathsf{sk_{id}} \leftarrow \mathsf{IBE.Extract}(\mathsf{msk}, \mathsf{id})$ *and* $\mathsf{negl}(\lambda)$ *is negligible in* λ*, and the probability is taken over the randomness of the algorithms.*

We recall the notions of ciphertext indistinguishability from random under adaptive chosen-identity and chosen plaintext attacks, denoted by INDr-ID-CPA which implies semantic security and recipient identity anonymity.

Definition 8 (INDr-ID-CPA Security). *Let* λ *be the security parameter. We say that an IBE system* IBE *is INDr-ID-CPA secure if for all p.p.t adversary* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$ *that makes queries to* OExt*,*

$$\mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{A}}(\lambda) := \left|\Pr\left[\mathsf{Exp}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{A}}(\lambda) = 1\right] - 1/2\right| \leq \mathsf{negl}(\lambda)$$

where the security experiment $\mathsf{Exp}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{A}}(\lambda)$*is defined in Fig. 3 and the probability is over the randomness of the* IBE *algorithms and the randomness of the attacker.*

An IBS system IBS with identity space $\mathcal{I}$ consists of four p.p.t algorithms. $\mathsf{IBS.Setup}(1^\lambda)$ returns a master public key pub and a master private key msk. We assume that pub is an implicit input to other algorithms. $\mathsf{IBS.Extract}(\mathsf{msk}, \mathsf{id})$. $\mathsf{IBS.Sign}(\mathsf{sk_{id}}, \mathsf{m})$ returns a signature s. $\mathsf{IBS.Ver}(\mathsf{id}, \mathsf{m}, s)$ returns 1 or 0. We define the correctness as follows.

Definition 9. *An IBS system* $\mathsf{IBS} = (\mathsf{IBS.Setup}, \mathsf{IBS.Extract}, \mathsf{IBS.Sign}, \mathsf{IBS.Ver})$ *is correct if for all* $(\mathsf{IBS.pub}, \mathsf{IBS.msk}) \leftarrow \mathsf{IBS.Setup}(1^\lambda)$*,* $\mathsf{sk_{id}} \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \mathsf{id})$*,*

$$\Pr\left[\mathsf{Ver}(\mathsf{id}, \mathsf{m}, \mathsf{Sign}(\mathsf{sk_{id}}, \mathsf{m}) = 1\right] \geq 1 - \mathsf{negl}(\lambda)$$

where $\mathsf{negl}(\lambda)$ *is negligible in* λ*, and the probability is over the algorithms.*

Experiment $\mathsf{Exp}^{\mathrm{indr}}_{\mathsf{IBE},\mathcal{A}}(\lambda)$:

1. $L_{\mathsf{id}} \leftarrow \emptyset$
2. $(\mathsf{pub}, \mathsf{msk}) \leftarrow \mathsf{Setup}(1^\lambda)$
3. $(\mathsf{m}*, \mathsf{id}^*, st) \leftarrow \mathcal{A}_1^{\mathsf{OExt}()}(\mathsf{pub})$
4. $b \leftarrow U(\{0,1\})$
5. $\mathsf{ct}_0^* \leftarrow \mathsf{IBE.Enc}(\mathsf{id}^*, \mathsf{m}^* b)$, $\mathsf{ct}_1^* \leftarrow \mathcal{C}_{\mathsf{ibe}}$
6. $b' \leftarrow \mathcal{A}_2^{\mathsf{OExt}()}(\mathsf{pub}, \mathsf{ct}_b^*, st)$
7. Return $(b = b') \wedge (\mathsf{id}^* \notin L_{\mathsf{id}})$

Oracle $\mathsf{OExt}(\mathsf{id})$:

1. Return $\mathsf{sk}_{\mathsf{id}} \leftarrow \mathsf{Extract}(\mathsf{msk}, \mathsf{id})$
2. $L_{\mathsf{id}} = \{\mathsf{id}\} \cup L_{\mathsf{id}}$

Experiment $\mathsf{Exp}^{\mathrm{euf}}_{\mathsf{IBS},\mathcal{A}}(\lambda)$:

1. $L_{\mathsf{id}} \leftarrow \emptyset$, $L_s \leftarrow \emptyset$
2. $(\mathsf{pub}, \mathsf{msk}) \leftarrow \mathsf{Setup}(1^\lambda)$
3. $(\mathsf{id}^*, \mathsf{m}^*, s^*) \leftarrow \mathcal{A}^{\mathsf{OExt}(),\mathsf{OSig}()}(\mathsf{pub})$
4. If $\mathsf{id}^* \in L_{\mathsf{id}}$, return 0
5. If $(\mathsf{id}^*, \mathsf{m}^*, s^*) \in L_s$, return 0
6. Return $\mathsf{Ver}(\mathsf{id}^*, \mathsf{m}^*, s^*)$

Oracle $\mathsf{OExt}(\mathsf{id})$:

1. Return $\mathsf{sk}_{\mathsf{id}} \leftarrow \mathsf{Extract}(\mathsf{msk}, \mathsf{id})$
2. $L_{\mathsf{id}} = \{\mathsf{id}\} \cup L_{\mathsf{id}}$

Oracle $\mathsf{OSig}(\mathsf{id}, \mathsf{m})$:

1. $\mathsf{sk}_{\mathsf{id}} \leftarrow \mathsf{Extract}(\mathsf{pub}, \mathsf{msk}, \mathsf{id})$
2. Return $s \leftarrow \mathsf{Sign}(\mathsf{sk}_{\mathsf{id}}, \mathsf{m})$
3. $L_s = \{\mathsf{id}, \mathsf{m}, s\} \cup L_s$

Fig. 3. INDr-ID-CPA and EUF-ID-CMA Security Experiments

We consider the standard unforgeability under chosen-identity and chosen-message attacks (EUF-ID-CMA). We also need a property called identity lossiness.

Definition 10. *Let λ be the security parameter. Consider the security experiment (game) defined in Fig. 3. We say an IBS system* IBS *is EUF-ID-CMA secure if for any p.p.t adversary $\mathcal{A}$ that makes polynomially many (in λ) queries the oracles* OExt() *and* OSig(),

$$\mathsf{Adv}^{\mathsf{euf}}_{\mathsf{IBS},\mathcal{A}}(\lambda) := \Pr[\mathsf{Exp}^{\mathsf{euf}}_{\mathsf{IBS},\mathcal{A}}(\lambda) = 1] \leq \mathsf{negl}(\lambda)$$

where the probability is over the randomness of $\mathcal{A}$ and the IBS system.

Definition 11. *We say that an IBS system* $\mathsf{IBS} = (\mathsf{IBS.Setup}, \mathsf{IBS.Extract}, \mathsf{IBS.Sign}, \mathsf{IBS.Ver})$ *with identity space $\mathcal{I}$ is η-identity lossy with respect to distribution $\mathcal{ID}$ over $\mathcal{I}$ if* $\tilde{\mathrm{H}}_\infty(\mathsf{id}|\mathsf{sk}_{\mathsf{id}}) \geq \mathrm{H}_\infty(\mathsf{id}) - \eta$ *for all* $(\mathsf{IBS.pub}, \mathsf{IBS.msk}) \leftarrow \mathsf{IBS.Setup}(1^\lambda)$, $\mathsf{sk}_{\mathsf{id}} \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \mathsf{id})$, $\mathsf{id} \leftarrow \mathcal{ID}$.

Appendix B Reduction for Proving Inequality (1)

We construct the reduction algorithm $\mathcal{B}$ that uses a distinguisher that distinguishes between Hyb_0 and Hyb_1 to break the INDr-ID-CPA security of the underlying IBE system IBE. We note that the proof works under the condition $\bar{\mathsf{M}}_1 : \forall \rho \in L_2, \rho \neq \mathsf{rcv}_0^* \wedge \rho \neq \mathsf{rcv}_1^*$. $\mathcal{B}$ is interacting with an INDr-ID-CPA challenger and works as follows.

$\mathcal{B}$ initialises three empty list L_1, L_2, and L_3. On receiving a master public key IBE.pub of the IBE system IBE. $\mathcal{B}$ sets $(\mathsf{IBS.pub}, \mathsf{IBS.msk}) \leftarrow \mathsf{IBS.Setup}(1^\lambda)$, and $\mathsf{pub} := (\mathsf{IBE.pub}, \mathsf{IBS.pub})$, $\mathsf{msk} := \mathsf{IBS.msk}$

Then, $\mathcal{B}$ responds to the IB-ME adversary $\mathcal{A}$'s queries as follows:

1. Query $\mathcal{O}_1(\sigma)$: $\mathcal{B}$ simply runs $\mathsf{sk}_\sigma \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \sigma)$, returns $\mathsf{ek}_\sigma = (\mathsf{sk}_\sigma, \sigma)$, and sets $L_1 = \{\sigma\} \sup L_1$.
2. Query $\mathcal{O}_2(\rho)$: $\mathcal{B}$ makes an OExt query on ρ to its IBE challenger, receives back $\mathsf{sk}_{\mathsf{id}}$ and sets $\mathsf{dk}_\rho := (\mathsf{sk}_\rho, \rho)$, and sets $L_2 = \{\sigma\} \cup L_2$.

Then, $\mathcal{A}$ announces $(\mathsf{m}_0^*, \mathsf{m}_1^*, \mathsf{rcv}_0^*, \mathsf{rcv}_1^*, \mathcal{ID}_0, \mathcal{ID}_1, st)$ and $\mathcal{B}$ creates the challenge ciphertext as follows:

1. For $i = 0, 1$, $\sigma_i^* \leftarrow \mathcal{ID}_i$, $\mathsf{ek}_{\sigma_i^*} \leftarrow \mathsf{IBS.Extract}(\mathsf{IBS.msk}, \sigma_i^*)$
2. $r^* \leftarrow U(\{0,1\}^\ell)$, $K_1^* \leftarrow U(\{0,1\}^s)$, $K_2^* \leftarrow U(\{0,1\}^v)$
3. $\kappa_1^* \leftarrow \mathsf{ReExt}(K_1^*, \sigma_0^*))$, $\kappa_2 \leftarrow \mathsf{Ext}(K_2^*, r^*)$
4. Send $(\mathsf{id}^* = \mathsf{rcv}_0^*, m^* := \mathsf{m}_0^* || r^*, st = \emptyset)$ to the challenger, receive back ct_b^* and set $\tilde{c}^* \leftarrow \mathsf{ct}_b^*$.
5. $\tilde{s} \leftarrow \mathsf{IBS.Sign}\,(\mathsf{sk}_\sigma, H(c^* || r^* || K_1^* || K_2^*))$, $s^* \leftarrow \tilde{s}^* \oplus \kappa_2^*$
6. Return $\mathsf{ct} := (c^*, s^*, K_1^*, K_2^*)$.

Then $\mathcal{A}$ makes oracle queries to $\mathcal{O}_1$, $\mathcal{O}_2$, $\mathcal{O}_{3,\beta}$. The $\mathcal{O}_1$ and $\mathcal{O}_2$ queries are answered in the same way as before. For the query $(\mathsf{rcv}, \mathsf{m})$ to $\mathcal{O}_{3,\beta}$, $\mathcal{B}$ returns $\mathsf{ct} = \mathsf{Enc}(\mathsf{ek}_{\sigma_\beta^*}, \mathsf{rcv}, \mathsf{m})$ where Enc is the encryption algorithm of the IB-ME construction. Finally, $\mathcal{B}$ outputs what $\mathcal{A}$ outputs.

We analyse the reduction. First of all, Hyb_0 and Hyb_1 have the same distribution on pub, which is correctly simulated by $\mathcal{B}$. Second, all $\mathcal{B}$ answers all queries properly. The only difference between Hyb_1 and Hyb_0 is the distribution of the value $\tilde{c}$. We can see that when the IBE challenger chose $b = 0$, i.e., the IBE challenge ciphertext ct^* is an encryption of IBE system, and $\mathcal{B}$ simulates Hyb_0. On the other hand, when the IBE challenger chose $b = 1$, ct^* is a random ciphertext, and $\mathcal{B}$ simulates Hyb_1. Let $\mathcal{B}\mathsf{Win}$ be the event that $\mathcal{B}$'s output equals to b. So,

$$\begin{aligned}
\Pr[\mathcal{B}\mathsf{Win}] &= \frac{1}{2}\left(\Pr[\mathcal{B}\mathsf{Win}|b=1] + \Pr[\mathcal{B}\mathsf{Win}|b=0]\right) \\
&= \frac{1}{2}\left(1 - \Pr[\mathcal{A} \to 0|b=1] + \Pr[\mathcal{A} \to 0|b=0]\right) \\
&= \frac{1}{2}\left(1 - \Pr[\mathsf{Hyb}_1 \Rightarrow 1] + \Pr[\mathsf{Hyb}_0 \Rightarrow 1]\right)
\end{aligned}$$

which gives $|\Pr[\mathsf{Hyb}_0 \Rightarrow 1] - \Pr[\mathsf{Hyb}_0 \Rightarrow 1]| \leq 2 \cdot \mathsf{Adv}^{\mathsf{indr}}_{\mathsf{IBE},\mathcal{A}}(\lambda)$ and ends the proof.

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